

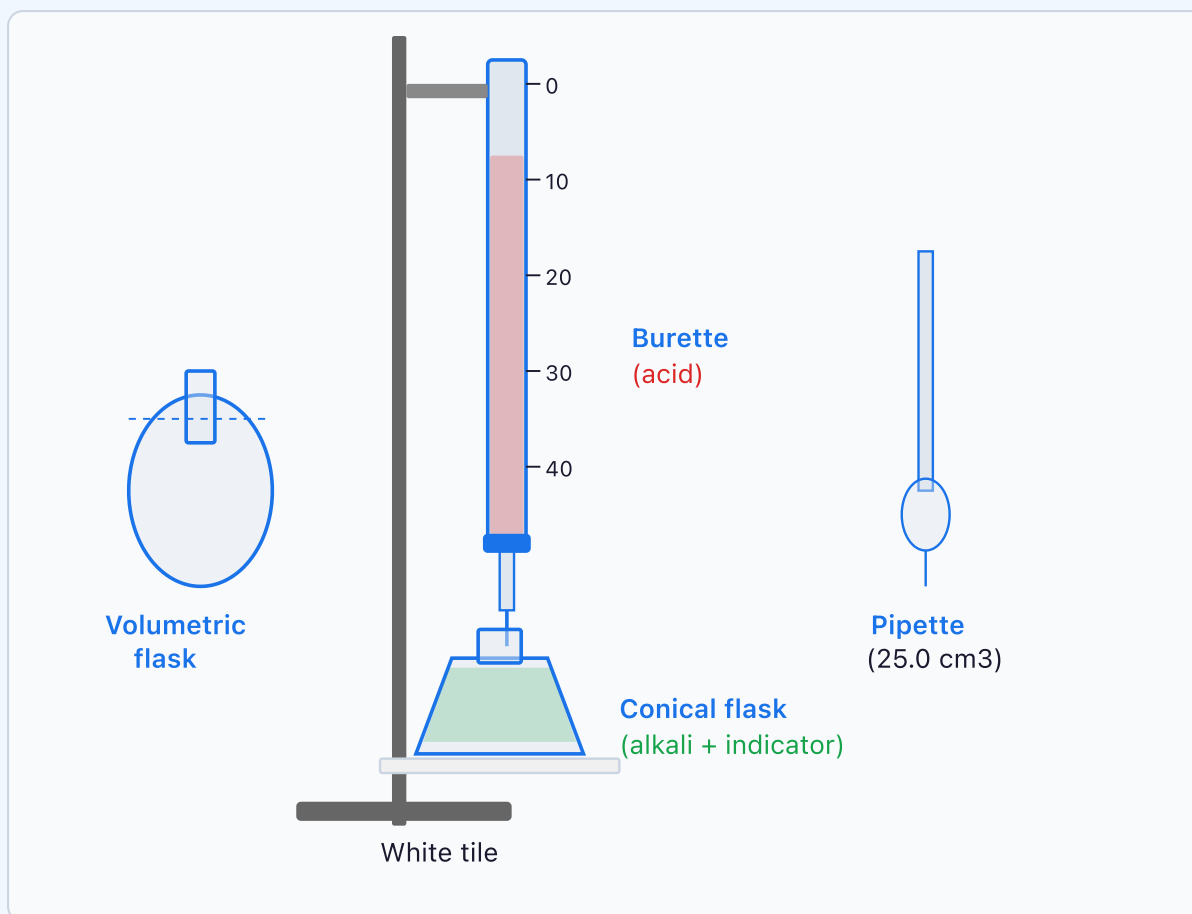
AQA A-Level Chemistry — Required Practicals

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RP1: Make Up a Volumetric Solution and Carry Out a Simple Acid-Base Titration



Background Theory

A **titration** is a quantitative technique used to determine the exact concentration of an unknown solution by reacting it with a solution of known concentration (the **standard solution**). In an acid-base titration, the equivalence point is reached when the moles of acid exactly equal the moles of base (for monoprotic acids and bases). An **indicator** changes colour at or near the equivalence point. The standard solution is prepared using a **volumetric flask** to ensure precise concentration. Calculations use: $n = c \times V$ (where n = moles, c = concentration in mol dm^{-3} , V = volume in dm^3), combined with the stoichiometric ratio from the balanced equation.

Apparatus

- **250 cm³ volumetric flask** with stopper (for preparing standard solution)
- **25.0 cm³ volumetric (bulb) pipette** (class B for routine work) and **pipette filler** (never mouth-pipette)
- **50 cm³ burette** and burette clamp on a clamp stand

- 250 cm³ conical flask (Erlenmeyer flask — shape allows swirling without spillage)
- White tile (placed under flask to see colour change more clearly)
- Indicator: methyl orange (strong acid/strong base, pH range 3.1–4.4) or phenolphthalein (weak acid/strong base, pH range 8.3–10.0)
- Balance (to 2 d.p., i.e. ± 0.01 g), wash bottle with distilled water, beakers, glass stirring rod, funnel

Method: Preparing a Standard Solution

1. Calculate the mass of solute needed for the desired concentration and volume using: $\text{mass} = n \times M_r = c \times V \times M_r$.
2. Weigh the solid accurately on a balance (record to **2 decimal places**). Transfer to a beaker.
3. Add a small volume (~50 cm³) of distilled water to the beaker. Stir with a glass rod to dissolve the solid completely. Gentle warming may be needed for slow-dissolving solids.
4. Allow the solution to cool to room temperature (volumetric flasks are calibrated at 20 degrees C). Pour the solution into the **250 cm³ volumetric flask** using a funnel.
5. **Rinse the beaker, stirring rod, and funnel several times** with small volumes of distilled water, adding all washings to the flask. This ensures complete transfer of all solute.
6. Add distilled water to just below the graduation mark. Then use a **dropping pipette** to add the final drops until the bottom of the **meniscus** sits exactly on the 250 cm³ line. (Read at eye level to avoid parallax.)
7. Stopper the flask and **invert 10–20 times** to mix thoroughly.

Method: Titration

1. **Rinse the pipette** with a small volume of the alkali solution (to prevent dilution from residual water). Then pipette exactly **25.0 cm³** of alkali into a clean conical flask.
2. Add **2–3 drops** of indicator.
3. **Rinse the burette** with a small volume of the acid solution. Fill the burette with acid and run some through the tap to remove air bubbles from the jet. Record the **initial burette reading** to the nearest **0.05 cm³** (half the smallest division).
4. **Rough titration:** Add acid from the burette to the conical flask while **swirling continuously**. When the indicator changes colour **permanently** (persists for >30 seconds), record the final

burette reading. Calculate the rough titre (this gives you an approximate end-point volume).

5. **Accurate titrations:** Repeat with fresh pipetted alkali. Add acid rapidly until $\sim 2 \text{ cm}^3$ before the expected end-point, then add **drop by drop** (or half-drops, by turning the tap to let a drop hang and washing it in with distilled water from a wash bottle). Record the final burette reading.
6. Repeat until you have at least **two concordant results** (within 0.10 cm^3 of each other).
7. Calculate the **mean titre** from concordant results only (exclude the rough and any anomalous results).

Key Variables

Variable	Detail
Independent	None (this is a quantitative determination, not a variable investigation)
Dependent	Titre volume (volume of acid needed to reach the end-point)
Control	Volume of alkali pipetted (exactly 25.0 cm^3), concentration of standard solution, indicator used, temperature

Safety

- Acids and alkalis are corrosive or irritant — wear **safety goggles** at all times. Gloves recommended.
- **Never mouth-pipette** — always use a pipette filler.
- Wipe up any spills immediately with plenty of water.
- Some indicators (e.g. phenolphthalein) are dissolved in ethanol — keep away from naked flames.

Calculations

$n = c \times V$ (V must be in dm^3 ; to convert from cm^3 , divide by 1000)

Use the **balanced equation** to find the mole ratio, then calculate the unknown

concentration:

$$c(\text{unknown}) = n(\text{unknown}) / V(\text{unknown})$$

Indicator Selection

Titration Type	pH at Equivalence	Suitable Indicator	Colour Change
Strong acid + strong base	7	Methyl orange or phenolphthalein (both work)	MO: red → yellow; PP: colourless → pink
Strong acid + weak base	<7	Methyl orange	Red → yellow
Weak acid + strong base	>7	Phenolphthalein	Colourless → pink
Weak acid + weak base	Variable	No suitable indicator (use pH meter)	—

Error Analysis

Source of Error	Type	Improvement
Misreading the burette scale (parallax)	Random	Read at eye level; read the bottom of the meniscus; take care with the 0.05 cm ³ estimate between scale markings
Overshooting the end-point	Systematic (makes titre too high)	Add acid drop-wise near the expected end-point; use half-drops; swirl between each addition
Incomplete transfer of solute to volumetric flask	Systematic (concentration too low)	Rinse the beaker and funnel at least 3 times with distilled water, adding all washings

Air bubble trapped in burette jet/tip	Systematic (titre appears larger than actual)	Run acid through the tap rapidly to expel air bubbles before starting; check the tip is full
Using the wrong indicator for the titration type	Systematic (wrong end-point)	Match indicator to the pH range at the equivalence point (see table above)
Pipette/burette not rinsed with the solution they will contain	Systematic (dilution)	Rinse the pipette with alkali and the burette with acid before use. Do NOT rinse the conical flask with alkali — it should be rinsed with distilled water only (extra alkali would require more acid)

Percentage Uncertainty

Burette: uncertainty = $\pm 0.05 \text{ cm}^3$ per reading, so $\pm 0.10 \text{ cm}^3$ for a titre (two readings).

% uncertainty = $(0.10 / \text{mean titre}) \times 100$. A larger titre gives a smaller % uncertainty.

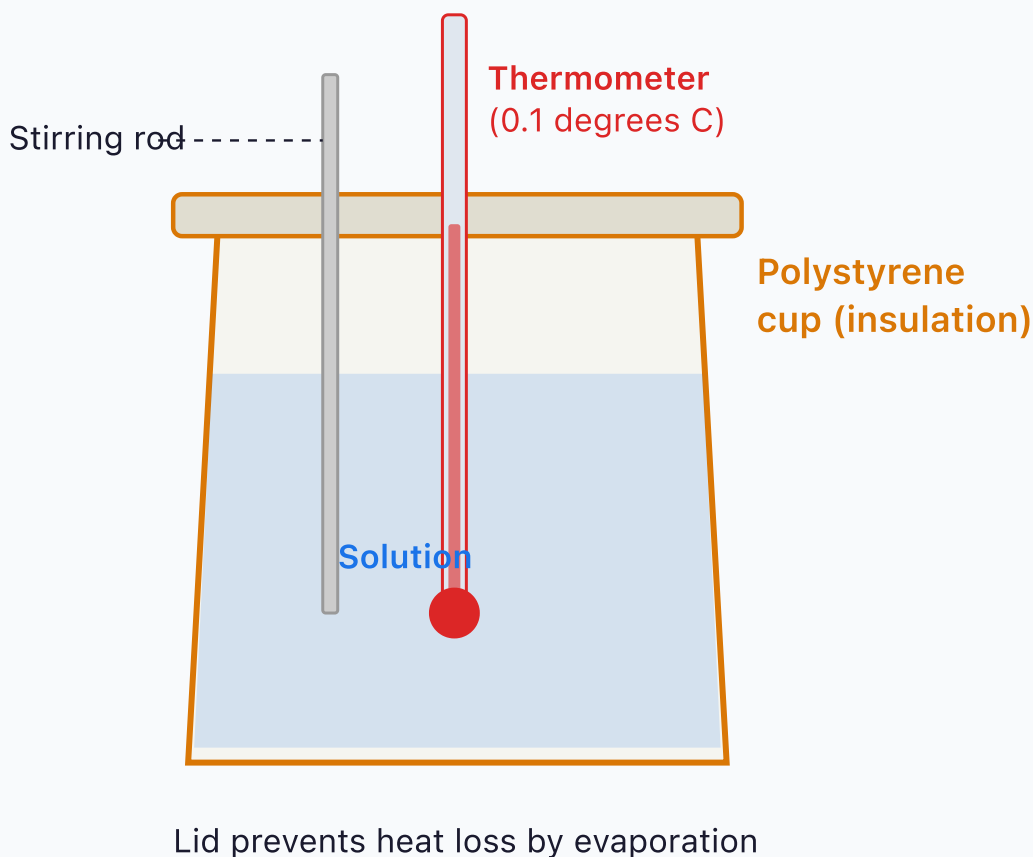
Pipette (25.0 cm³): uncertainty = $\pm 0.06 \text{ cm}^3$. % uncertainty = $(0.06 / 25.0) \times 100 = 0.24\%$.

Balance (2 d.p.): uncertainty = $\pm 0.01 \text{ g}$ per reading, so $\pm 0.02 \text{ g}$ for a mass by difference.

Overall % uncertainty: add individual percentage uncertainties.

Exam Tip: Always quote burette readings to **2 decimal places** (e.g. 23.45 cm^3), even if the second decimal is 0 (e.g. 23.40 , not 23.4). Concordant means within **0.10 cm^3** . Only average concordant titres — always exclude the rough. When rinsing: rinse the pipette with the solution it will contain; rinse the conical flask with distilled water only (not the solution). This is a very common exam question — explain why for each.

RP2: Measurement of an Enthalpy Change (Calorimetry)



Background Theory

Enthalpy changes (ΔH) are measured using **calorimetry**: the heat released or absorbed by a reaction is transferred to a known mass of water (or solution), and the temperature change is measured. The heat energy transferred is calculated using **$q = mc\Delta T$** , where m = mass of solution (g), c = specific heat capacity ($4.18 \text{ J g}^{-1} \text{ K}^{-1}$ for dilute aqueous solutions), and ΔT = temperature change (K or degrees C). The enthalpy change per mole is then $\Delta H = -q / n$, where n = moles of the limiting reagent. The sign is negative for exothermic reactions (temperature rises) and positive for endothermic (temperature falls).

A **polystyrene cup** is used as a simple calorimeter because polystyrene is an excellent thermal insulator (low thermal conductivity), reducing heat loss to the surroundings. A lid prevents heat loss by evaporation. Despite these precautions, some heat is always lost, so the measured ΔT is less than the true value, and the experimental ΔH is always less exothermic (or less endothermic) than the true value.

Apparatus

- Polystyrene cup (nested in a beaker for stability) with a cardboard or polystyrene lid
- Thermometer (± 0.1 degrees C) or calibrated temperature probe with data logger
- Measuring cylinder (25 or 50 cm^3) for solutions, or balance for solids
- Stirring rod (glass or plastic), stopwatch

Method (Enthalpy of Neutralisation)

1. Measure **25.0 cm^3** of one solution (e.g. 1.0 mol dm^{-3} HCl) using a measuring cylinder and pour into the polystyrene cup.
2. Record the temperature of the solution **every minute for 3 minutes** to establish a stable baseline (pre-mixing trend).
3. At the **4th minute**, quickly add 25.0 cm^3 of the second solution (e.g. 1.0 mol dm^{-3} NaOH). Stir gently and replace the lid.
4. Continue recording the temperature **every minute for at least 6–8 more minutes** (post-mixing cooling trend).
5. Plot temperature (y-axis) against time (x-axis). Draw two lines of best fit: one for the pre-mixing readings (usually flat or slightly sloping) and one for the post-mixing readings (usually decreasing as heat is lost to surroundings).
6. **Extrapolate** both lines back to the time of mixing (minute 4). The vertical distance between the two extrapolated lines at the mixing time gives the **corrected ΔT** . This graphical extrapolation method compensates for heat loss.
7. Calculate: $q = mc\Delta T$ (where m = total mass of solution, assuming density = 1 g cm^{-3}).
8. Calculate: $\Delta H = -q / n$ (n = moles of limiting reagent). Express in kJ mol^{-1} .

Safety

- Acids and alkalis are corrosive — wear safety goggles and gloves.
- Take care with strongly exothermic reactions — the cup may become hot.
- If using powdered metals or displacement reactions, the mixture can splash — use a lid.

Key Assumptions and Limitations

- Specific heat capacity of the solution = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ (same as pure water). In reality, dissolved salts slightly alter c
- Density of the solution = 1.0 g cm^{-3} (very close to true for dilute solutions)
- No heat is absorbed by the calorimeter itself (polystyrene has very low heat capacity, so this is a reasonable approximation)
- All heat from the reaction is transferred to the solution (some is lost to surroundings despite insulation)

Error Analysis

Source of Error	Type	Effect on ΔH	Improvement
Heat loss to surroundings	Systematic	Measured ΔT too small; ΔH underestimated (less exothermic)	Use a lid, insulation, and the graphical extrapolation technique. Or use a bomb calorimeter (commercial)
Assuming $c = 4.18$ and density = 1.0	Systematic	Introduces a small systematic error in q	Acknowledge the assumption; use more accurate values if known
Thermometer slow to respond (thermal lag)	Random	May miss the peak temperature	Use a fast-response temperature probe with data logger; stir well
Incomplete reaction	Systematic	ΔT too small; ΔH underestimated	Use an excess of one reagent; stir thoroughly; allow sufficient time
Heat absorbed by the calorimeter	Systematic	ΔT of solution too small	Determine the calorimeter constant; use a more insulated system

Exam Tip: The **graphical extrapolation** method is essential for correcting heat loss. Draw lines of best fit for the pre- and post-mixing data, then extrapolate both to the time of

mixing to find the true ΔT . Always state the two key assumptions: $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$ and density = 1.0 g cm^{-3} . Experimental ΔH values are always less exothermic (smaller magnitude) than literature values due to heat loss — if asked why, say "heat was lost to the surroundings, so the measured temperature change was smaller than the true value."

RP3: How the Rate of a Reaction Changes with Temperature (Sodium Thiosulfate + HCl)

Background Theory

The reaction between sodium thiosulfate and hydrochloric acid produces sulfur as a precipitate, which makes the solution cloudy:



A cross drawn on paper beneath the flask becomes obscured as sulfur forms. The time taken for the cross to disappear is measured. A shorter time indicates a faster rate. Rate is proportional to $1/t$. According to collision theory, increasing temperature increases the kinetic energy of particles, leading to (1) more frequent collisions and (2) a greater proportion of collisions exceeding the activation energy (E_a). The second factor is dominant: the Boltzmann distribution shifts so that many more particles have energy $\geq E_a$, dramatically increasing the rate. A 10 degrees C rise typically approximately **doubles** the rate.

Apparatus

- Conical flask (100 cm^3), measuring cylinders (10, 25 cm^3)
- 0.1 mol dm^{-3} sodium thiosulfate solution, 1.0 mol dm^{-3} dilute HCl
- Thermometer (± 1 degrees C), water baths at 20, 30, 40, 50, 60 degrees C
- Paper with a bold black cross (X) drawn on it, stopwatch

Method

1. Measure **25 cm^3** of sodium thiosulfate into a conical flask.

2. Place the flask in a water bath at the target temperature. Allow to **equilibrate for 5 minutes**.
3. Record the exact temperature of the thiosulfate solution.
4. Place the flask on the paper with the cross. Add **5 cm³** of dilute HCl and **immediately start the stopwatch**.
5. Look down through the flask from above. Stop the timer when the cross is **no longer visible** due to the sulfur precipitate making the solution opaque.
6. Record the time (t). Calculate **rate = 1/t**.
7. Repeat at five different temperatures (e.g. 20, 30, 40, 50, 60 degrees C). Use fresh solutions each time.

Safety

- **SO₂ gas** is produced (toxic, respiratory irritant) — work in a well-ventilated area or fume cupboard.
- Dilute HCl is an irritant — wear safety goggles.
- Take care with hot water baths at higher temperatures — risk of scalding.

Analysis

- Plot **1/t (y-axis)** against **temperature in degrees C (x-axis)**. The graph shows an exponential-shaped increase, confirming that rate increases with temperature.
- For an **Arrhenius plot**: plot **ln(1/t) (y-axis)** against **1/T (x-axis)**, where T is in Kelvin. This gives a straight line with gradient = $-E_a/R$. Calculate E_a from the gradient: $E_a = -\text{gradient} \times R$ (where $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$).

Error Analysis

Source of Error	Type	Improvement
Subjective judgement of when the cross disappears	Random	The same person should judge every time. Better: use a light sensor or colorimeter placed below the flask for an objective, reproducible end-point

Temperature changes during the reaction	Systematic	Use a thermostatically controlled water bath; keep the flask in the bath during the reaction
Human reaction time in starting/stopping the stopwatch	Random	Use data logging; repeat each temperature at least 3 times and take the mean
Volumes measured with measuring cylinder (low precision)	Random	Use a pipette for more accurate volumes; alternatively, ensure consistent technique

Exam Tip: Rate = $1/t$ is a **proxy for initial rate** (the same amount of product — enough sulfur to obscure the cross — is formed in each experiment). A 10 degrees C rise approximately doubles the rate. For the Arrhenius plot, remember to convert temperature to **Kelvin** ($T = \text{degrees C} + 273$). The gradient = $-E_a/R$. Always state units for rate: $1/t$ has units of **s^{-1}** .

RP4: Test-Tube Reactions to Identify Cations and Anions

Background Theory

Qualitative analysis identifies unknown ions through characteristic reactions: **flame tests** (metal cations emit characteristic wavelengths of light when electrons are excited and then return to ground state), **precipitation reactions** with NaOH or NH_3 (insoluble metal hydroxides form with characteristic colours), and **confirmatory tests** for anions (Cl^- , Br^- , I^- , SO_4^{2-} , CO_3^{2-}).

Cation Identification

Cation	Flame Colour	NaOH (few drops)	NaOH (excess)	NH_3 (few drops)	NH_3 (excess)

Li^+	Crimson red	—	—	—	—
Na^+	Strong yellow	—	—	—	—
K^+	Lilac	—	—	—	—
Ca^{2+}	Orange-red	Slight white ppt	Slight white ppt	No ppt	No ppt
Sr^{2+}	Red	Slight white ppt	Slight white ppt	No ppt	No ppt
Ba^{2+}	Green	No ppt	No ppt	No ppt	No ppt
NH_4^+	—	No precipitate. Warm with NaOH(aq): NH_3 gas is given off, which turns damp red litmus paper blue . $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$			
Cu^{2+}	Blue-green	Blue ppt	Insoluble	Blue ppt	Dissolves: deep blue solution $[\text{Cu}(\text{NH}_3)_4]^{2+}$
Fe^{2+}	—	Green ppt	Insoluble	Green ppt	Insoluble
Fe^{3+}	—	Brown ppt	Insoluble	Brown ppt	Insoluble
Al^{3+}	—	White ppt	Dissolves (amphoteric: $\text{Al}(\text{OH})_4^-$)	White ppt	Insoluble
Mg^{2+}	—	White ppt	Insoluble	White ppt	Insoluble

Anion Identification

Anion	Test	Positive Result	Equation
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OH^-	Dip red litmus paper into the solution (or use universal indicator)	Red litmus turns blue (pH well above 7)	—
CO_3^{2-}	Add dilute HCl; pass gas through limewater	Effervescence; limewater turns milky	$\text{CO}_3^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CO}_2$; $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
SO_4^{2-}	Acidify with dilute HNO_3 , then add $\text{BaCl}_2(\text{aq})$	White precipitate (BaSO_4)	$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
Cl^-	Acidify with dilute HNO_3 , then add $\text{AgNO}_3(\text{aq})$	White precipitate (AgCl)	$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
Br^-	Acidify with dilute HNO_3 , then add $\text{AgNO}_3(\text{aq})$	Cream precipitate (AgBr)	$\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$
I^-	Acidify with dilute HNO_3 , then add $\text{AgNO}_3(\text{aq})$	Yellow precipitate (AgI)	$\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$

Important Details

- **Flame tests:** Clean the nichrome wire by dipping in concentrated HCl and heating until no colour is seen. Repeat until clean. For potassium (lilac), the yellow sodium flame can mask it — view through **blue cobalt glass** which absorbs the sodium yellow.
- **Acidify before adding BaCl_2 or AgNO_3 :** Dilute HNO_3 removes carbonates and sulfites that could give false positive precipitates (BaCO_3 and Ag_2CO_3 would also be white precipitates).
- **Group 2 cations (the AQA focus of RP4):** solubility of the hydroxides **increases** down the group, so Mg^{2+} gives a white precipitate with NaOH, Ca^{2+} and Sr^{2+} give only a slight precipitate, and Ba^{2+} gives none. Solubility of the sulfates **decreases** down the group, so adding dilute H_2SO_4 gives no precipitate with Mg^{2+} , a slight one with Ca^{2+} , and a white precipitate with Sr^{2+} and Ba^{2+} . The Fe, Cu and Al rows above are covered in detail in RP11.

- **Al³⁺ vs Mg²⁺ distinction:** Both give white precipitates with NaOH and NH₃. The key difference: Al(OH)₃ dissolves in **excess NaOH** (amphoteric) but Mg(OH)₂ does not. This is the classic distinguishing test.
- **Cu²⁺ with excess NH₃:** The blue Cu(OH)₂ precipitate dissolves to give a **deep blue solution** of the tetraammine complex [Cu(NH₃)₄(H₂O)₂]²⁺. This is a ligand substitution reaction.

Safety

- Silver nitrate (AgNO₃) stains skin permanently — wear gloves.
- Barium chloride (BaCl₂) is toxic — handle with care; wash hands thoroughly.
- Concentrated HCl is corrosive and produces fumes — use in a fume cupboard for flame tests.
- Wear safety goggles throughout all test-tube reactions.

Error Analysis

Source of Error	Type	Improvement
Contaminated nichrome wire (carries ions from previous test)	Systematic	Clean the wire thoroughly in HCl and flame until no colour before each test
Na ⁺ contamination giving persistent yellow flame	Systematic	Na ⁺ is ubiquitous (in glass, skin oils); view through blue cobalt glass to filter out the yellow
Not acidifying before adding BaCl ₂ or AgNO ₃	Systematic	Always acidify with dilute HNO ₃ first to remove interfering anions (CO ₃ ²⁻ , SO ₃ ²⁻)
Difficulty distinguishing cream from white precipitate	Random	Compare directly side-by-side with a known Cl ⁻ standard (white AgCl)

Exam Tip: The key distinguishing test between Al³⁺ and Mg²⁺ is excess NaOH: Al(OH)₃ is **amphoteric** and dissolves (Al(OH)₃ + OH⁻ → Al(OH)₄⁻), while Mg(OH)₂ does not. Cu²⁺

dissolves in excess NH_3 (ligand substitution: $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 4\text{H}_2\text{O}$) but NOT in excess NaOH . These two facts are tested very frequently.

RP5: Distillation of a Product from a Reaction

Background Theory

Distillation separates liquids based on differences in boiling point. In organic synthesis, it is used to purify the desired product from a reaction mixture. The mixture is heated; the component with the lowest boiling point evaporates first, passes through a **condenser** (where it is cooled and re-liquefied), and is collected separately. Monitoring the **thermometer** at the still head allows collection of a specific fraction at the correct boiling point. A pure liquid has a **sharp, constant boiling point**; an impure mixture boils over a range.

Reflux is a related technique where the vapour is condensed and returned to the flask rather than collected — this allows prolonged heating without losing volatile reagents or products, ensuring the reaction goes to completion before distillation.

Apparatus

- Round-bottomed flask (appropriate size for the volume of liquid), still head with thermometer adapter
- Thermometer (range covering the expected boiling point), **Liebig condenser** with rubber tubing
- Collecting flask or conical flask, **anti-bumping granules** (added before heating)
- Heat source (Bunsen burner with gauze, electric heating mantle, or water bath for low-boiling-point liquids)
- Clamp stand, boss, clamps (flask and condenser both clamped for safety)

Method

1. Set up the distillation apparatus: round-bottomed flask → still head (with thermometer, bulb level with the side-arm outlet) → condenser (angled slightly downward) → collecting flask.

2. Add **anti-bumping granules** to the flask before heating. Never add them to a hot liquid (risk of violent boiling).
3. Pour the reaction mixture into the flask (no more than **half full** to prevent bumping over).
4. Connect the condenser: cold water enters at the **bottom** and exits at the **top** (counter-current flow for maximum cooling efficiency).
5. Turn on the water supply. Then begin heating gently, increasing heat gradually.
6. Discard the first few drops of distillate (**forerun**) which may contain impurities.
7. Collect the fraction that distils within **± 2 degrees C** of the expected boiling point of the desired product.
8. Stop collecting when the temperature starts to rise above the target range (the next component is beginning to distil).

Safety

- **Never heat a sealed system** — the collecting end must be open to the atmosphere to prevent dangerous pressure build-up.
- Many organic liquids are **highly flammable** — use an electric heating mantle or water bath rather than a naked flame.
- Anti-bumping granules prevent **superheating** and sudden, violent boiling.
- Ensure all glassware joints are secure but not forced (forced joints can crack).

Analysis

- Record the boiling range of the collected fraction. A **sharp boiling point** (± 1 degrees C) indicates high purity.
- Calculate **percentage yield**: (actual moles of product / theoretical moles) $\times 100$.
- Percentage yield is always $< 100\%$ due to: incomplete reaction, side reactions, mechanical losses (product left in flask, condenser, transfer losses), and losses during purification.

Error Analysis

Source of Error	Type	Improvement
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Product left in flask, condenser, and connecting glassware	Systematic (lowers yield)	Rinse apparatus with small portions of solvent (adds to yield but may reduce purity)
Collecting fractions outside the correct temperature range	Systematic (reduces purity)	Monitor the thermometer carefully; only collect at the correct boiling point ± 2 degrees C
Overheating causing decomposition	Systematic	Heat gently and gradually; use a water bath for thermally sensitive compounds
Incomplete reaction before distillation	Systematic (lower yield)	Ensure the reaction has gone to completion — reflux first if necessary

Exam Tip: Water flows **counter-current** in the condenser (in at the bottom, out at the top) for maximum cooling. The thermometer bulb should be level with the **side-arm outlet** (to measure the temperature of the vapour entering the condenser, not the liquid in the flask). Anti-bumping granules must be added **before** heating. Percentage yield is always $<100\%$ — state reasons (incomplete reaction, side reactions, transfer losses, purification losses).

RP6: Tests for Functional Groups (Alcohol, Aldehyde, Alkene, Carboxylic Acid)

Background Theory

Organic functional groups can be identified using characteristic chemical tests. Each test exploits a specific reaction of the functional group, producing an observable change (colour change, precipitate, gas evolution). These tests are essential for identifying unknown organic compounds and for confirming that a synthesis has produced the expected product.

Summary of Tests

Functional Group	Test Reagent	Conditions	Positive Result	Chemistry
Alkene (C=C)	Bromine water (Br ₂ (aq))	Room temperature; shake	Orange/brown decolourises	Electrophilic addition: C=C + Br ₂ → dibromo compound
Aldehyde (–CHO)	Tollens' reagent ([Ag(NH ₃) ₂] ⁺)	Warm gently in water bath (~60 degrees C)	Silver mirror forms on clean glass tube	Ag ⁺ reduced to Ag(s); aldehyde oxidised to carboxylic acid
Aldehyde (–CHO)	Fehling's solution (Cu ²⁺ in alkaline tartrate)	Heat in water bath	Blue → brick-red precipitate (Cu ₂ O)	Cu ²⁺ reduced to Cu ⁺ ; aldehyde oxidised
Carboxylic acid (–COOH)	Na ₂ CO ₃ solution	Room temperature	Effervescence (CO ₂ gas); confirm with limewater	2RCOOH + Na ₂ CO ₃ → 2RCOONa + H ₂ O + CO ₂
Primary/secondary alcohol	Acidified K ₂ Cr ₂ O ₇ (orange)	Warm gently	Orange → green (Cr ³⁺)	Alcohol oxidised (1 degree → aldehyde → carboxylic acid; 2 degree → ketone). Tertiary: no reaction
Tertiary alcohol	Acidified K ₂ Cr ₂ O ₇	Warm	No change (stays orange)	Cannot be oxidised (no H on C–OH carbon)

Safety

- Tollens' reagent can form explosive **silver nitride (Ag_3N)** on standing — always prepare fresh, use immediately, and wash away promptly after the test
- Bromine water is toxic, corrosive, and produces irritating vapour — use in a fume cupboard; wear gloves
- Potassium dichromate(VI) is a **known carcinogen and mutagen** — handle with gloves; avoid skin contact and inhalation of dust
- Organic unknowns may be flammable — keep away from naked flames

Distinguishing 1 degree, 2 degree, and 3 degree Alcohols

Alcohol Type	With $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$	Product	Tollens'/Fehling's on Product
Primary (1 degree)	Orange → green	Aldehyde (then carboxylic acid if excess oxidant)	Positive (silver mirror / red ppt)
Secondary (2 degree)	Orange → green	Ketone	Negative (ketones not oxidised)
Tertiary (3 degree)	No colour change	No reaction	N/A

Exam Tip: Tollens' and Fehling's test for **aldehydes but NOT ketones**. This is because aldehydes can be further oxidised to carboxylic acids (the Ag^+ or Cu^{2+} acts as the oxidising agent), but ketones cannot be oxidised without breaking C–C bonds. The oxidation of 1 degree alcohols produces aldehydes (if distilled immediately) or carboxylic acids (if heated under reflux with excess oxidant). To distinguish 1 degree from 2 degree alcohols: oxidise both, then test the products with Tollens'/Fehling's — only the product from 1 degree (aldehyde) will give a positive result.

RP7 (Part A): Measuring the Rate of Reaction by an Initial Rate Method (Iodine Clock)

Background Theory

The **iodine clock** uses a two-stage reaction system to measure the initial rate of reaction. In the first (slow) step, iodine (I_2) is gradually produced by the reaction of iodate and bisulfite ions. In the second (fast) step, the I_2 is immediately consumed by reacting with the remaining hydrogensulfite (bisulfite) ions. When all the hydrogensulfite has been used up, free I_2 accumulates and reacts with starch indicator to give a **sudden blue-black colour**. The time to this colour change is inversely proportional to the initial rate. By varying the concentration of one reactant while keeping others constant (and maintaining a constant total volume with distilled water), you can determine the **order of reaction** with respect to that reactant.

Apparatus

- Conical flasks or beakers, measuring cylinders (multiple, to avoid contamination)
- Stopwatch, white tile (to see colour change), starch indicator solution
- Solutions: potassium iodate (KIO_3), sodium hydrogensulfite (bisulfite, $NaHSO_3$)
- Distilled water (to keep total volume constant)

Method

1. Prepare a series of mixtures with varying concentration of **one reactant** (e.g. KIO_3). **Keep total volume constant** by adding distilled water to make up the difference.
2. Add starch indicator to each flask.
3. Add the second reagent, start the stopwatch immediately, and swirl to mix.
4. Record the time (t) for the solution to turn **dark blue-black**.
5. Calculate **rate = $1/t$** for each experiment.
6. Plot rate ($1/t$, y-axis) against concentration (x-axis).

Determining Order of Reaction

Graph Shape (1/t vs [X])	Order	Interpretation
Straight line through origin	First order	Rate is directly proportional to [X]: Rate = $k[X]$
Curve (parabolic shape through origin)	Second order	Rate is proportional to $[X]^2$: Rate = $k[X]^2$. Confirm by plotting rate vs $[X]^2$ — should give a straight line
Horizontal line	Zero order	Rate is independent of [X]: Rate = k

Why Total Volume Must Be Constant

If the total volume changes between experiments, then changing the volume of one reactant also changes the effective concentrations of all other reactants in the mixture. By adding distilled water to compensate, you ensure that only the concentration of the **one variable reactant** changes while everything else remains constant. This is essential for a valid investigation.

Safety

- Solutions are generally low hazard at the concentrations used, but wear goggles.
- Iodine can stain skin and clothing — handle carefully.

Error Analysis

Source of Error	Type	Improvement
Subjective end-point (when is the solution "blue-black enough"?)	Random	Same person judges every time; use a colorimeter for objective detection
Temperature variation between experiments	Systematic	Use a thermostatted water bath; record temperature for each run
Delay in starting timer after mixing	Random	One person mixes, another starts the timer; use data logging

Measuring cylinder uncertainty
(volumes not precise enough)

Random

Use burettes or pipettes for more
accurate volumes

Exam Tip: The iodine clock effectively measures the **time for a fixed small amount of reaction** to occur (the amount of I_2 that reacts with the hydrogensulfite). Since this is a small fraction of the total reaction, it approximates the **initial rate**. Always keep total volume constant by adding water when reducing a reactant volume. If asked to determine the order from data: compare experiments where one concentration doubles — if the rate doubles, it is first order; if the rate quadruples, it is second order.

RP7 (Part B): Measuring the Rate of Reaction by a Continuous Monitoring Method (Gas Collection or Mass Loss)

Background Theory

In **continuous monitoring**, the progress of a reaction is tracked over time by measuring either (a) the **volume of gas produced** (using a gas syringe) or (b) the **mass lost** (as gas escapes from an open flask on a balance). A typical reaction is the decomposition of calcium carbonate by hydrochloric acid: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$. Plotting volume or mass change against time gives a curve from which the **rate at any point** can be determined from the gradient of a tangent. The **initial rate** is the gradient at $t = 0$.

Apparatus

- Conical flask with bung and delivery tube connected to a **gas syringe** (100 cm^3), OR conical flask with cotton wool plug on a **top-pan balance** (to $\pm 0.01\text{ g}$)
- Stopwatch, measuring cylinders for reagent volumes
- Reagents: marble chips ($CaCO_3$) + dilute HCl (1 or 2 mol dm^{-3})

Method

1. Measure the required volume of HCl into the conical flask.
2. Weigh the marble chips (use the same mass each time for valid comparisons).
3. Add the marble chips, quickly fit the bung (gas syringe method) or replace the cotton wool plug (mass loss method), and start the stopwatch.
4. Record gas volume (or mass) at regular intervals (every 30 seconds) until the reaction is complete (no further gas produced / mass constant).
5. Repeat with a changed variable: different HCl concentration, different particle size (powdered vs large chips), or different temperature.

Analysis

- Plot **volume of gas (or mass lost)** on y-axis vs **time** on x-axis.
- The **initial rate** = gradient of the tangent to the curve at $t = 0$ (steeper = faster).
- The curve levels off when one reactant is completely consumed. The **final volume/mass change** depends on which reactant is in excess.
- Compare curves for different conditions: higher concentration → steeper initial gradient and faster levelling off; same final amount if excess acid is used.

Effect of Variables on Rate

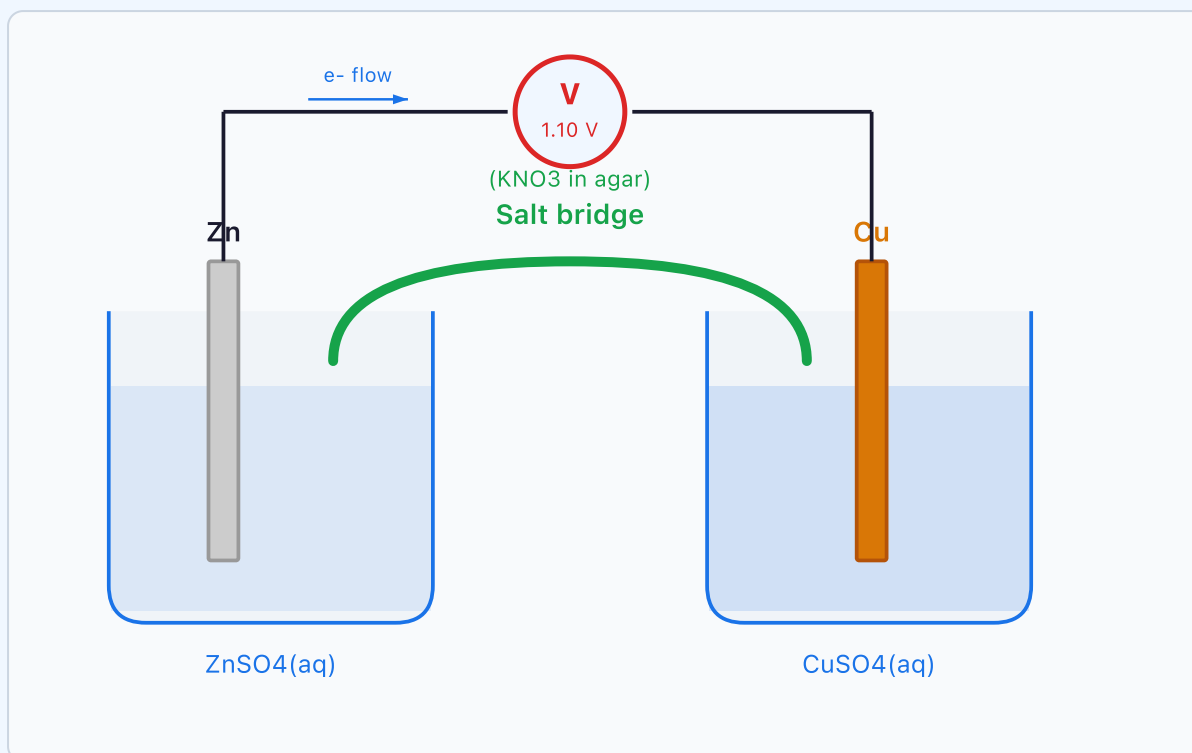
Variable Changed	Effect on Rate	Explanation (Collision Theory)
Increased concentration	Faster	More particles per unit volume → more frequent collisions → more successful collisions per second
Smaller particle size (powder vs chips)	Faster	Greater surface area exposed → more collisions per second between acid and solid surface
Higher temperature	Faster	Particles have more kinetic energy → more frequent collisions AND a greater proportion with energy $\geq E_a$
Catalyst added	Faster	Provides an alternative pathway with a lower activation energy → greater proportion of collisions are successful

Error Analysis

Source of Error	Type	Improvement
Gas escapes before bung is fitted	Systematic (total volume too low)	Fit the bung as quickly as possible; use a side-arm flask and add acid via syringe through a septum
Gas syringe sticking due to friction	Systematic (volume too low)	Lubricate the syringe; tap gently; use a new syringe
CO ₂ dissolves in the aqueous solution	Systematic (volume measured is less than produced)	Acknowledged limitation; the gas syringe collects only what escapes the solution
Uneven marble chip sizes between experiments	Random	Use the same total mass; crush to approximately the same size; weigh accurately

Exam Tip: On a volume-time graph: the **steeper the initial gradient**, the faster the initial rate. If excess acid is used, all curves reach the **same final volume** (same moles of CaCO₃ reacted). If excess CaCO₃ is used, the final volume depends on the moles of acid. Always state which is in excess and why. The **cotton wool plug** (mass loss method) allows gas to escape but prevents acid spray from leaving the flask.

RP8: Measuring the EMF of an Electrochemical Cell



Background Theory

An **electrochemical cell** converts chemical energy into electrical energy through spontaneous redox reactions. It consists of two **half-cells**, each containing a metal electrode immersed in a solution of its ions. The more reactive metal (more negative E°) acts as the **anode** (oxidation: $\text{M} \rightarrow \text{M}^{n+} + n\text{e}^-$), and the less reactive metal acts as the **cathode** (reduction: $\text{M}^{n+} + n\text{e}^- \rightarrow \text{M}$). Electrons flow from anode to cathode through the external wire. The **salt bridge** (e.g. KNO_3 solution in agar) completes the circuit by allowing ion flow between the half-cells, maintaining electrical neutrality.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

Apparatus

- Two beakers (250 cm^3), metal electrodes (e.g. Zn strip, Cu strip, Fe strip, Mg ribbon)
- 1.0 mol dm^{-3} solutions of the corresponding metal salts (ZnSO_4 , CuSO_4 , FeSO_4 , MgSO_4)
- Salt bridge: filter paper soaked in KNO_3 solution, or agar- KNO_3 gel in a U-tube

- **High-resistance voltmeter** (or multimeter set to DC voltage) — high resistance minimises current flow and gives a more accurate EMF reading
- Connecting wires with crocodile clips, emery paper (to clean electrodes)

Method

1. **Clean** both metal electrodes with emery paper to remove the oxide layer (oxide increases resistance and lowers the reading).
2. Pour $1.0 \text{ mol dm}^{-3} \text{ ZnSO}_4$ into one beaker and $1.0 \text{ mol dm}^{-3} \text{ CuSO}_4$ into the other.
3. Place the Zn electrode in the ZnSO_4 and the Cu electrode in the CuSO_4 .
4. Connect the salt bridge (dip one end in each beaker).
5. Connect electrodes to the voltmeter using wires. Record the **EMF**.
6. For Zn/Cu: expected $E^\circ_{\text{cell}} = +0.34 - (-0.76) = \mathbf{+1.10 \text{ V}}$.
7. Repeat with different metal combinations (Zn/Fe, Fe/Cu, Mg/Cu, etc.) and compare measured EMFs with calculated values.

Safety

- CuSO_4 is harmful if ingested — avoid skin contact; wear gloves.
- Wear safety goggles throughout.
- Dispose of heavy metal solutions according to local regulations (do not pour down the drain).

Error Analysis

Source of Error	Type	Improvement
Oxide layer on electrode surface	Systematic (lowers EMF)	Clean with emery paper immediately before use
Concentrations not exactly 1.0 mol dm^{-3}	Systematic	Prepare solutions accurately using a volumetric flask

Salt bridge drying out or making poor contact	Systematic (resistance increases, EMF drops)	Ensure the salt bridge is freshly prepared and fully immersed in both solutions
Voltmeter drawing current	Systematic (lowers reading)	Use a high-resistance voltmeter or potentiometer to minimise current flow
Temperature not 298 K (25 degrees C)	Systematic	Record temperature; acknowledge deviation from standard conditions; use Nernst equation for correction

Exam Tip: The more reactive metal (more negative E^0) is always the **anode** (oxidation). Electrons flow from the anode to the cathode through the external wire. The **salt bridge** allows ion flow to complete the internal circuit and maintain electrical neutrality (anions migrate towards the anode, cations towards the cathode). $E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}}$. A positive cell EMF means the reaction is spontaneous (feasible). KNO_3 is used in the salt bridge because K^+ and NO_3^- ions have similar mobilities and do not react with the half-cell solutions.

RP9: pH Changes When a Weak Acid Reacts with a Strong Base and a Strong Acid Reacts with a Weak Base

Background Theory

This practical produces a **pH titration curve** that reveals key features of the acid-base system. For a weak acid (e.g. ethanoic acid, CH_3COOH) titrated with a strong base (e.g. NaOH), the curve has distinctive features: (1) the initial pH is above 1 (weak acid is partially dissociated), (2) there is a **buffer region** where pH changes slowly (mixture of weak acid and its conjugate base resists pH change), (3) at the **half-equivalence point**, exactly half the acid has been neutralised, so $[\text{HA}] = [\text{A}^-]$ and **pH = pK_a**, (4) the equivalence point has **pH > 7** (the salt of a weak acid and

strong base is basic due to hydrolysis of the conjugate base), (5) use **phenolphthalein** as indicator (changes at pH 8.3–10.0, matching the equivalence pH).

Apparatus

- Calibrated pH meter or pH probe with data logger
- 50 cm³ burette filled with 0.10 mol dm⁻³ NaOH
- Beaker (250 cm³) containing 25.0 cm³ of 0.10 mol dm⁻³ CH₃COOH
- Magnetic stirrer and stirrer bar, clamp stand, wash bottle with distilled water

Method

1. **Calibrate the pH meter** using at least two buffer solutions (e.g. pH 4 and pH 7, or pH 7 and pH 10).
2. Pipette 25.0 cm³ of weak acid into the beaker. Place on the magnetic stirrer.
3. Immerse the pH probe. Record the initial pH.
4. Fill the burette with NaOH. Record the initial burette reading.
5. Add NaOH in **1 cm³ increments** initially. After each addition: stir, wait 10–15 seconds for a stable reading, then record pH and total volume added.
6. When approaching the equivalence point (rapid pH change), switch to **0.5 cm³ or even 0.1 cm³ increments** to capture the steep region accurately.
7. Continue well past the equivalence point until pH levels off (typically pH 12–13).
8. Plot **pH (y-axis) against volume of NaOH added (x-axis)**.

Key Features of the Titration Curve

Feature	Detail
Initial pH	~2.9 for 0.10 mol dm ⁻³ CH ₃ COOH (higher than a strong acid of the same concentration because it is only partially dissociated)
Buffer region	Flat section before equivalence: mixture of CH ₃ COOH and CH ₃ COO ⁻ acts as a buffer, resisting pH change. pH changes slowly per cm ³ of NaOH added

Half-equivalence point	Half the acid neutralised: $[\text{CH}_3\text{COOH}] = [\text{CH}_3\text{COO}^-]$, so pH = pK_a (for ethanoic acid, pK _a = 4.76). Read directly from the graph
Equivalence point	pH > 7 (~8.7 for this system). The solution contains sodium ethanoate (CH_3COONa); the ethanoate ion is a base that undergoes hydrolysis: $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$
Post-equivalence	pH levels off at ~12–13 (excess NaOH dominates)

Second Titration: Strong Acid with a Weak Base

The specification requires both combinations. Repeat the method with 25.0 cm³ of 0.10 mol dm⁻³ ammonia solution in the beaker and 0.10 mol dm⁻³ hydrochloric acid in the burette. Record pH after each addition, using small increments near the equivalence point, and plot pH against volume of acid added.

Feature	Detail
Initial pH	About 11 for 0.10 mol dm ⁻³ NH ₃ (lower than a strong base of the same concentration because ammonia is only partially protonated)
Buffer region	Mixture of NH ₃ and NH ₄ ⁺ resists pH change before the equivalence point
Equivalence point	pH < 7 (about 5 for this system): the solution contains NH ₄ Cl, and the ammonium ion is a weak acid: $\text{NH}_4^+ \rightleftharpoons \text{NH}_3 + \text{H}^+$
After equivalence	pH falls towards about 1–2 as excess HCl dominates
Indicator	The steep section lies at acidic pH, so methyl orange is suitable; phenolphthalein is not

Safety

- NaOH is corrosive — wear goggles and gloves.
- Weak acids are irritants — avoid skin contact.
- Ammonia solution releases an irritating vapour — use in a well-ventilated room.
- pH probes are fragile and expensive — handle with care; store in the correct storage solution.

Error Analysis

Source of Error	Type	Improvement
pH meter not calibrated or poorly calibrated	Systematic	Calibrate with at least two (ideally three) buffer solutions spanning the expected pH range
Adding too much NaOH near the equivalence point	Systematic (misses the steep region)	Use smaller increments (0.1–0.2 cm ³) around the equivalence point
Not allowing pH to stabilise before reading	Random	Wait 10–15 seconds after each addition; ensure thorough mixing with the stirrer
CO ₂ absorption from the atmosphere into the NaOH	Systematic (concentration of NaOH slightly lower than expected)	Use freshly prepared NaOH; keep the beaker covered; store NaOH in a sealed container

Exam Tip: Be ready to sketch and compare both curves: strong acid/weak base has its equivalence point below pH 7 (use methyl orange), weak acid/strong base above pH 7. For a weak acid/strong base titration: equivalence pH > 7, so use **phenolphthalein** (not methyl orange). At the half-equivalence point: $[HA] = [A^-]$, so $pH = pK_a$. This is the easiest way to determine pK_a experimentally. Calculate $K_a = 10^{-pK_a}$. The buffer region demonstrates the Henderson-Hasselbalch equation: $pH = pK_a + \log([A^-]/[HA])$.

RP10: Preparation of a Pure Organic Solid (with Purity Test) and a Pure Organic Liquid

Background Theory

This required practical has two parts. In **Part A** an organic solid is made, separated and purified by **recrystallisation**, and its purity is checked by measuring its **melting point**. A typical example is aspirin, made by reacting 2-hydroxybenzoic acid with ethanoic anhydride (a few drops of concentrated acid catalyse the reaction). In **Part B** an organic liquid is made and purified. A typical example is cyclohexene, made by dehydrating cyclohexanol with concentrated phosphoric(V) acid; the product is distilled off, washed in a separating funnel, dried and redistilled.

A pure solid melts sharply at a temperature matching the data-book value. Impurities **lower** the melting point and make it melt over a **wider range**. A pure liquid distils at a constant temperature equal to its boiling point.

Apparatus

- **Solid:** conical flask, water bath, Buchner funnel and flask with filter paper (vacuum filtration), ice bath, hot water for recrystallisation, melting-point apparatus with capillary tubes
- **Liquid:** round-bottomed or pear-shaped flask, distillation apparatus (still head, thermometer, Liebig condenser), **separating funnel**, anhydrous drying agent (e.g. anhydrous calcium chloride or magnesium sulfate), anti-bumping granules
- Balance, measuring cylinders, glass rod, sample tubes

Method Part A: Purifying an Organic Solid (Recrystallisation)

1. Carry out the reaction and collect the crude solid by **vacuum filtration** using a Buchner funnel.
2. Dissolve the crude solid in the **minimum volume of hot solvent**. The product should be very soluble in the hot solvent but only slightly soluble when cold.
3. If insoluble impurities are present, filter the hot solution quickly through a warmed funnel.

4. Let the filtrate cool slowly, then stand it in ice. Crystals of the product form, while soluble impurities stay in solution.
5. Collect the crystals by vacuum filtration and wash them with a **small volume of ice-cold solvent**.
6. Dry the crystals between filter papers or in a warm oven, then weigh them.

Method Part A: Testing Purity by Melting Point

1. Seal one end of a capillary tube. Push a little of the dry, powdered solid into it to a depth of 2–3 mm.
2. Place the tube in the melting-point apparatus next to the thermometer and heat **slowly** (about 1–2 degrees C per minute) near the expected melting point.
3. Record the temperature at which the solid starts to melt and the temperature at which it has completely melted.
4. Compare the range with the data-book value. A narrow range (within about 1–2 degrees C) at the expected value indicates a pure product.

Method Part B: Preparing and Purifying an Organic Liquid

1. Mix the reagents in a flask with anti-bumping granules and heat. Distil off the crude product (for cyclohexene, collect the distillate below about 90 degrees C).
2. Transfer the distillate to a **separating funnel**. Wash it with water or sodium carbonate solution to remove acidic and water-soluble impurities. Release the pressure regularly by inverting the funnel and opening the tap.
3. Allow the layers to separate, run off and discard the lower (aqueous) layer, and keep the organic layer. Check which layer is which from the densities.
4. Run the organic layer into a dry flask and add a few lumps of an **anhydrous drying agent**. Swirl and stopper until the liquid is clear rather than cloudy.
5. Decant or filter the dried liquid into a clean flask and **redistil**, collecting the fraction that boils within a few degrees of the product's boiling point (cyclohexene boils at about 83 degrees C).

Safety

- Concentrated sulfuric and phosphoric(V) acids are **corrosive**; wear goggles and gloves.
- Ethanoic anhydride is corrosive and its vapour is irritating; use a fume cupboard.

- Organic products and solvents are often **flammable**; heat with a water bath or electric heater, not a naked flame.
- Always vent the separating funnel so pressure does not build up.

Analysis

- Calculate the **percentage yield** = (actual mass / theoretical mass) x 100, using the limiting reagent.
- Losses happen at every stage: the reaction may be incomplete, some product stays dissolved in the recrystallisation solvent, and some is lost during transfers and filtration.
- A low, broad melting range shows the solid is still impure (or still wet); recrystallise it again.

Error Analysis

Source of Error	Type	Improvement
Too much solvent used in recrystallisation	Systematic (lower yield)	Use the minimum volume of hot solvent; add it in small portions
Crystals still damp when weighed	Systematic (yield too high, melting point too low)	Dry to constant mass before weighing and before the melting-point test
Heating the melting-point apparatus too quickly	Systematic (melting point reads too high)	Heat slowly near the expected value; repeat with a fresh sample
Water left in the organic liquid	Systematic (impure product, cloudy)	Leave with drying agent until clear; redistil

Exam Tip: Know the purpose of every step. Minimum hot solvent: so the solution becomes saturated and as much product as possible crystallises on cooling. Filtering while hot: removes insoluble impurities. Cooling in ice: lowers solubility so more crystals form. Washing with ice-cold solvent: removes soluble impurities left on the surface without dissolving much product. Sodium carbonate wash: removes acid (look for effervescence).

Anhydrous salt: removes water, and the liquid turns from cloudy to clear. An impure solid has a **lower and broader** melting range.

RP11: Test-Tube Reactions to Identify Transition Metal Ions in Aqueous Solution

Background Theory

In water, metal ions exist as **metal-aqua ions** such as $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. Adding a base removes H^+ from the water ligands, giving neutral, insoluble hydroxide precipitates with characteristic colours. Some precipitates then react further: with excess ammonia, **ligand substitution** can give a soluble ammine complex, and with excess sodium hydroxide an **amphoteric** hydroxide dissolves. The AQA specification covers Fe^{2+} , Cu^{2+} , Fe^{3+} and Al^{3+} (Al^{3+} is not a transition metal ion but is studied alongside them) with OH^- , NH_3 and CO_3^{2-} .

The 3+ aqua ions are **more acidic** than the 2+ ions because the smaller, more highly charged ion polarises the O–H bonds in the water ligands more strongly. This is why M^{3+} ions give CO_2 effervescence with sodium carbonate, while M^{2+} ions give metal carbonate precipitates.

Apparatus

- Test tubes and rack, dropping pipettes
- Dilute solutions (about 0.1 mol dm^{-3}) of copper(II) sulfate, iron(II) sulfate (freshly made), iron(III) chloride or nitrate, aluminium sulfate
- Dilute $\text{NaOH}(\text{aq})$, dilute $\text{NH}_3(\text{aq})$, $\text{Na}_2\text{CO}_3(\text{aq})$

Method

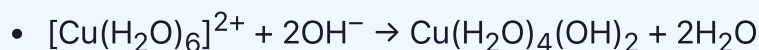
1. Put about 1 cm^3 of the metal ion solution into a test tube.
2. Add $\text{NaOH}(\text{aq})$ **dropwise** and record the colour of any precipitate. Then add NaOH **in excess**, shake, and record any change.

3. Repeat with fresh samples using $\text{NH}_3(\text{aq})$, first dropwise and then in excess.
4. Repeat with fresh samples using $\text{Na}_2\text{CO}_3(\text{aq})$, noting both precipitates and any effervescence.
5. Use the pattern of results to identify an unknown ion.

Results

Ion	Aqua ion colour	NaOH (few drops / excess)	NH_3 (few drops / excess)	Na_2CO_3
Cu^{2+}	Blue	Blue ppt $\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2$ / no further change	Blue ppt / dissolves to a deep blue solution $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$	Blue-green ppt CuCO_3
Fe^{2+}	Pale green	Green ppt $\text{Fe}(\text{H}_2\text{O})_4(\text{OH})_2$, turns brown on standing in air / no further change	Green ppt / no further change	Green ppt FeCO_3
Fe^{3+}	Yellow-brown (violet when fully hydrated)	Brown ppt $\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3$ / no further change	Brown ppt / no further change	Brown ppt $\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3$ and effervescence (CO_2)
Al^{3+}	Colourless	White ppt $\text{Al}(\text{H}_2\text{O})_3(\text{OH})_3$ / dissolves to give colourless $[\text{Al}(\text{OH})_4]^-$	White ppt / no further change	White ppt $\text{Al}(\text{H}_2\text{O})_3(\text{OH})_3$ and effervescence (CO_2)

Key Equations



- $\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2 + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 2\text{H}_2\text{O} + 2\text{OH}^-$
- $2[\text{Fe}(\text{H}_2\text{O})_6]^{3+} + 3\text{CO}_3^{2-} \rightarrow 2\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3 + 3\text{CO}_2 + 3\text{H}_2\text{O}$
- $\text{Al}(\text{H}_2\text{O})_3(\text{OH})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4]^- + 3\text{H}_2\text{O}$

Safety

- Sodium hydroxide is corrosive to eyes and skin; wear goggles.
- Ammonia solution gives off an irritating vapour; use in a well-ventilated room.
- Copper(II) salts are harmful; avoid skin contact and wash hands afterwards.

Error Analysis

Source of Error	Type	Improvement
Fe^{2+} solution partly oxidised to Fe^{3+}	Systematic (precipitate looks brownish)	Use freshly prepared iron(II) sulfate solution and observe the precipitate straight away
Not adding enough reagent to see the "excess" result	Random	Add the reagent until the tube is well over half full and shake
Judging colours in poor light	Random	Hold tubes against a white background; compare with known samples

Exam Tip: Two results come up again and again: Cu^{2+} is the only ion here that dissolves in **excess ammonia** (deep blue solution), and Al^{3+} is the only one that dissolves in **excess NaOH** (amphoteric). With sodium carbonate, M^{2+} ions give MCO_3 precipitates, but M^{3+} ions give the hydroxide precipitate plus CO_2 bubbles, because the 3+ aqua ions are acidic enough to react with carbonate. Always include the correct formula of each precipitate, with water ligands, in equations.

RP12: Separation of Species by Thin-Layer Chromatography (TLC)

Background Theory

In **thin-layer chromatography**, the **stationary phase** is a thin layer of silica or alumina on a glass, metal or plastic plate, and the **mobile phase** is a solvent. The solvent rises up the plate by capillary action. Each substance in the sample is distributed between the two phases: species that are **adsorbed more strongly** onto the stationary phase, or are less soluble in the solvent, move more slowly. Different substances therefore travel different distances and separate into spots.

$$R_f = \text{distance moved by the spot} / \text{distance moved by the solvent front}$$

Under the same conditions (same stationary phase, solvent and temperature), a substance always has the same R_f value, so spots can be identified by comparing them with reference samples run on the same plate.

Apparatus

- TLC plate, pencil and ruler, capillary tubes for spotting
- Beaker or TLC tank with a lid (or watch glass), suitable solvent
- Samples to be analysed (for example, a mixture of dyes, analgesics, or amino acids) and known reference compounds
- Means of locating colourless spots: UV lamp (for plates with a fluorescent indicator), iodine crystals, or ninhydrin spray for amino acids

Method

1. Wearing gloves and holding the plate by its edges, draw a faint **pencil** baseline about 1 cm from the bottom.
2. Use a capillary tube to put a small, concentrated spot of each sample and each reference on the baseline, leaving gaps between them. Let the spots dry.

3. Pour solvent into the beaker to a depth **below the baseline** and cover it so the air inside becomes saturated with solvent vapour.
4. Stand the plate in the beaker and replace the lid. Leave it until the solvent has risen most of the way up the plate.
5. Remove the plate and immediately mark the **solvent front** in pencil. Let it dry in a fume cupboard.
6. Locate colourless spots (UV lamp, iodine or ninhydrin) and draw round them in pencil.
7. Measure the distance from the baseline to the centre of each spot and to the solvent front, and calculate R_f values.

Safety

- Many TLC solvents are **flammable and harmful**; use small volumes in a fume cupboard, away from flames.
- Iodine vapour and ninhydrin are harmful; handle in a fume cupboard and wear gloves.
- Do not look directly at the UV lamp; it can damage the eyes.

Analysis

- Compare the R_f values and positions of the sample spots with those of the references run on the same plate.
- A pure substance gives a **single spot**; a mixture gives several spots.
- If two substances have similar R_f values in one solvent, they may separate better in a different solvent.

Error Analysis

Source of Error	Type	Improvement
Spots too large, so they spread and overlap	Random	Apply small spots, drying between applications to build up concentration
Solvent front not marked straight away	Systematic (R_f values too large)	Mark the front as soon as the plate is removed, before the solvent evaporates

Beaker not covered	Random (uneven solvent front)	Keep the lid on so the atmosphere is saturated with solvent vapour
Touching the plate surface	Systematic (contamination)	Wear gloves and hold the plate by the edges

Exam Tip: Common "why" questions: the baseline is drawn in **pencil** because ink would dissolve in the solvent and move up the plate; the solvent must be **below the baseline** or the samples would dissolve into the solvent pool; the lid keeps the atmosphere saturated so the solvent does not evaporate from the plate. Separation depends on the balance between **adsorption** to the stationary phase and **solubility** in the mobile phase. R_f has no units and is always less than 1.

Supporting Content (Not AQA Required Practicals)

The two sections below are not on AQA's list of 12 required practicals. They are kept because they support related content: standard electrode potentials (background to RP8) and Le Chatelier's principle.

Supporting Content: Standard Electrode Potentials (Using the SHE)

Background Theory

Standard electrode potential ($E^\ominus$) is the EMF of a half-cell measured against the **Standard Hydrogen Electrode (SHE)** under standard conditions: all solutions at **1.0 mol dm^{-3}** , gases at **1 atm (100 kPa)**, temperature at **298 K (25 degrees C)**. The SHE is assigned $E^\ominus = 0.00 \text{ V}$ by convention. It consists of a platinised platinum electrode in 1.0 mol dm^{-3} HCl with H_2 gas bubbled over it at 1 atm. The platinum is inert (does not react) but provides a surface for the equilibrium: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$.

In practice, the SHE is awkward to use (H_2 gas supply needed). A **secondary reference electrode** (e.g. Ag/AgCl or calomel) with a known, stable E° is often used instead, and results are converted to the SHE scale.

Method

1. Set up the SHE: platinised Pt electrode in 1.0 mol dm^{-3} HCl, H_2 bubbled at 1 atm, at 298 K.
2. Set up the test half-cell: metal electrode in 1.0 mol dm^{-3} solution of its ions (or Pt electrode for non-metal half-cells, e.g. $\text{Fe}^{3+}/\text{Fe}^{2+}$).
3. Connect with a salt bridge (KNO_3). Connect both to a high-resistance voltmeter.
4. Read the EMF and its sign. This is the E° of the test half-cell (since $E^\circ(\text{SHE}) = 0.00 \text{ V}$).
5. Repeat for multiple half-cells to build an electrochemical series.

Standard Conditions Summary

Condition	Standard Value
Temperature	298 K (25 degrees C)
Solution concentration	1.0 mol dm^{-3}
Gas pressure	100 kPa
SHE reference	$E^\circ = 0.00 \text{ V}$ by definition

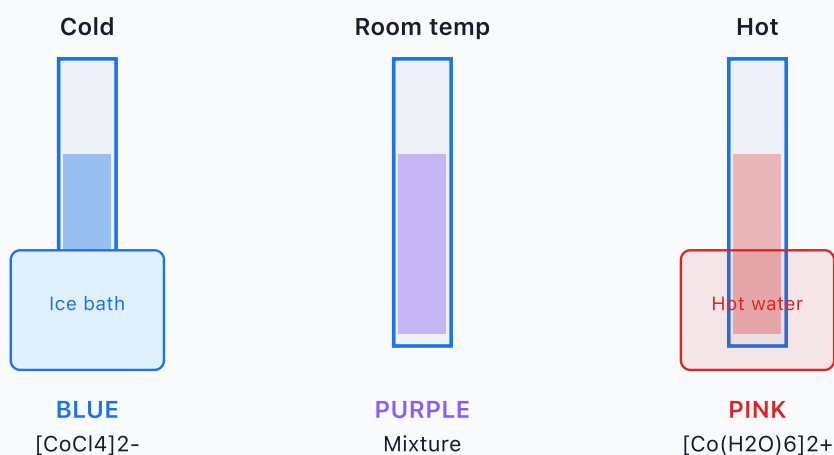
Safety

- H_2 gas is **flammable and explosive** — no naked flames; well-ventilated area.
- HCl is corrosive — goggles and gloves.
- Metal salt solutions may be toxic — handle with care; wash hands.

Exam Tip: Standard conditions: **298 K, 100 kPa, 1.0 mol dm^{-3}** . The SHE is assigned 0.00 V by **convention** (not because it actually has zero potential). For non-metal half-cells (e.g. Cl_2/Cl^- , $\text{Fe}^{3+}/\text{Fe}^{2+}$), use a **platinum electrode** (inert conductor). A more negative E° means

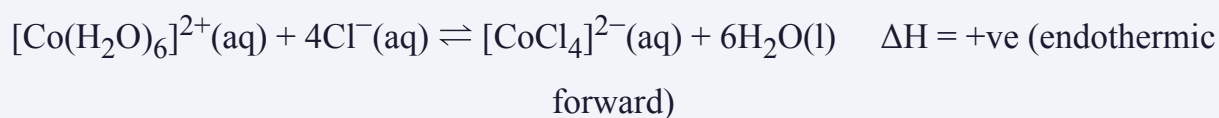
the species is a stronger reducing agent (more easily oxidised). A more positive E^0 means a stronger oxidising agent (more easily reduced).

Supporting Content: Effect of Changing Conditions on a Reversible Reaction at Equilibrium



Background Theory

Le Chatelier's principle: When a system at equilibrium is subjected to a change in concentration, temperature, or pressure, the equilibrium shifts in the direction that **opposes the change**, partially counteracting it. This practical uses the cobalt(II) chloride equilibrium, which has a visible colour change between the pink aqueous complex $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and the blue tetrahedral complex $[\text{CoCl}_4]^{2-}$.



Apparatus

- Test tubes and rack, cobalt(II) chloride solution in water/HCl mixture (starting purple)
- Concentrated HCl (source of Cl^-), distilled water, test pipettes
- Ice bath (~0 degrees C) and hot water bath (~80 degrees C)

Method and Predictions

Change Applied	Le Chatelier's Prediction	Direction of Shift	Colour Change
Heating (hot water bath)	System absorbs heat by shifting in the endothermic direction	Shifts right (forward)	Purple → blue (more $[\text{CoCl}_4]^{2-}$)
Cooling (ice bath)	System releases heat by shifting in the exothermic direction	Shifts left (backward)	Purple → pink (more $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$)
Adding concentrated HCl (more Cl^-)	System shifts to remove added Cl^-	Shifts right	Purple → blue
Adding distilled water (dilutes Cl^-)	System shifts to produce more aqueous ions (replace removed Cl^-)	Shifts left	Purple → pink

Important Concepts

- **Temperature changes** alter the **value of K_c** . Heating an endothermic reaction increases K_c (more products at equilibrium); cooling decreases K_c .
- **Concentration changes** shift the position of equilibrium but do **NOT change K_c** (K_c only changes with temperature).
- **A catalyst** does NOT shift the position of equilibrium and does NOT change K_c . It increases the rate of both the forward and reverse reactions equally, so equilibrium is reached **faster** but at the same position.

Safety

- Cobalt(II) chloride is **toxic and a suspected carcinogen** — wear gloves and avoid skin contact.
- Concentrated HCl is **corrosive** and produces fumes — use in a fume cupboard.
- Hot water bath — handle with care to avoid burns.

Error Analysis

Source of Error	Type	Improvement
Subjective colour judgement (when is it "blue" vs "purple"?)	Random	Use a colorimeter to measure absorbance objectively; compare with standard solutions
Temperature of water bath not constant	Systematic	Monitor with a thermometer; add ice or heat as needed to maintain temperature
Equilibrium not fully re-established before observing	Systematic	Allow sufficient time at each temperature/concentration before recording the colour

Exam Tip: Le Chatelier's principle: the system shifts to **oppose** the imposed change. For temperature: heating favours the **endothermic** direction; cooling favours the **exothermic** direction. For concentration: adding a species shifts equilibrium **away** from that species; removing it shifts **towards** it. A **catalyst does NOT shift equilibrium** — it only speeds up attainment. Temperature is the **ONLY** factor that changes the value of K_c . This distinction (position vs value of K_c) is heavily tested.